Table 12—Heterogeneity in the Effect on Borrowing, Attainment, and Financial Outcomes by Great Recession Severity

Panel A. Borrowing, attainment and earnings outcomes

Dependent variable:	Cumulative loans		Attainment and earnings, 10 years after entry				
	4 years after entry (1)	6 years after entry (2)	Years enrolled (3)	Credits attempted (4)	Any degree (5)	BA degree (6)	ln(earnings) (7)
Constrained × *cohort* $\in \{2006, 2007, 2008\}$	1,954	2,054	0.10	5.3	0.044	0.052	0.047
	(541)	(581)	(0.03)	(1.1)	(0.011)	(0.011)	(0.017)
	{0.017}	{0.029}	{0.004}	{0.002}	{0.004}	{0.003}	{0.021}
× Δ*UR* 2007–2009	1,220	1,419	−0.01	0.2	−0.011	−0.008	−0.030
	(732)	(852)	(0.05)	(1.6)	(0.015)	(0.016)	(0.031)
	{0.276}	{0.187}	{0.715}	{0.841}	{0.135}	{0.425}	{0.094}

Panel B. Borrowing and financial outcomes

Dependent variable:	Cumulative loans		Financial outcomes, 10 years after entry				
	4 years after entry (8)	6 years after entry (9)	Ever delinquent (stud. loans) (10)	Ever default (stud. loans) (11)	Any delinquent debt (12)	Any mortgage (13)	Any auto loan (14)
Constrained × *cohort* $\in \{2006, 2007, 2008\}$	1,978	3,085	−0.013	−0.018	−0.001	0.000	−0.009
	(382)	(558)	(0.005)	(0.005)	(0.004)	(0.006)	(0.006)
	{0.865}	{0.626}	{0.486}	{0.832}	{0.295}	{0.530}	{0.479}
× Δ*UR* 2007–2009	−40	−208	−0.001	0.000	−0.004	−0.002	−0.002
	(206)	(333)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)
	{0.867}	{0.645}	{0.467}	{0.821}	{0.254}	{0.519}	{0.491}

Notes: Texas four-year entrant sample (panel A) or CCP/Equifax sample (panel B). Dependent variable is listed in the column headings. Each panel contains results from separate regressions of the outcome on an indicator for being constrained at entry interacted with the percentage point change in the county unemployment rate where a student first enrolled in college (panel A) or is first observed in the data (panel B). Regressions also include standard controls and interactions between control variables and the change in the unemployment rate. The change in unemployment rate variable is normalized to have a within-sample mean of zero. Robust standard errors, clustered by entry institution (Texas sample) or entry state (CCP sample), in parentheses; *p*-values from wild cluster bootstrap-*t* in brackets.

way we address this concern is by restricting our sample to students who first enrolled prior to the start of the Great Recession, thereby mitigating the likelihood of an effect of the economic downturn on enrollment. As another check, we test for heterogeneity in the estimated effects of loan limit increases by the economic conditions in a student's home county in the year before and year of college entry. We estimate models in which our main specification is fully interacted with indicators for above- and below-median annual county unemployment. We find very little in the way of heterogeneous effects for four-year entrants (online Appendix Tables C.34 and C.35), suggesting that student responses to the policy change do not appear to vary based on the intensity of exposure to the recession prior to enrollment.

Second, we test for heterogeneous effects for students who were (potentially) differentially impacted by the Great Recession while in college. If constrained students were more or less affected by the recession than unconstrained students, we would expect our estimates to vary with the severity of the recession. To measure recession severity, we focus on the economic conditions in the county in which a student's college is located (or is first observed in the credit data) and adapt the approach developed in Yagan (2019) to our setting. Specifically, we calculate the change in the average unemployment rate between 2007 and 2009 in the county in which a student first enrolled in college (or is first observed in the credit data)